

The question I am answering is: *do stocks held by the same large institutional investors tend to move together more during periods of market stress?* The project looks at large U.S. firms and models institutional ownership as a network, where firms are nodes which are connected if they share the same institutional owners. The goal is to see whether these ownership connections are related to how strongly stock returns move together when markets experience large shocks.

The data collection has been slightly changed since the proposal.

1. I planned to scrape Yahoo to extract stock price data, but instead used it to establish the list of the largest publicly traded companies on US exchanges.
  - a. Link: <https://finance.yahoo.com/>
2. I collected market volatility data through the FRED API, because it is free and easy to access. The FRED data has been stored in stress\_days.csv.
  - a. <https://fred.stlouisfed.org/>
3. I will obtain daily stock return data and institutional ownership data based on SEC Form 13F filings using the Python interface to WRDS. WRDS does not allow web scraping so it should be done this way.
  - a. <https://wrds-www.wharton.upenn.edu/>

The main focus of the analysis is the same as in my proposal: to understand whether institutional ownership networks are associated with stronger stock return comovement during periods of market stress. Market stress is defined using realized market volatility based on historic S&P 500 returns (FRED data).

The analysis has three main steps.

1. The first step of the analysis is to identify periods of market stress using realized market volatility. I used daily S&P 500 prices collected from FRED to calculate daily returns and rolling volatility. Days with unusually high volatility (90th percentile) are classified as stress days, and consecutive stress days may be grouped into longer stress periods. At the moment, I have daily volatility data saved in a CSV file, and I am deciding whether to analyze stress at the daily level or to use multi day stress periods. I will decide based on which approach shows clearer patterns and enough observations for correlation analysis.
2. I will use institutional ownership data to build an ownership network across firms. Each firm represents a node, and two firms are connected if they share one or more institutional investors. The presence of a connection indicates whether two firms share any institutional owners, while the strength of the connection reflects how many institutional investors the firms share. I will calculate degree centrality, which measures how many other firms a firm is connected to through shared institutional ownership, as well as weighted degree, which shows how strongly a firm is connected by accounting for the total amount of ownership overlap across all of its connections.
3. I will calculate pairwise return correlations for all firm pairs during the identified stress periods. These correlations measure how closely two firms' stock prices move together during stress. I will then compare correlations across firm pairs with different levels of ownership overlap and examine whether firms with higher network centrality tend to have stronger average comovement with other firms. The analysis will be descriptive and

exploratory, and I will look at correlation comparisons, and simple regression models (CAPM).

I will also include the following visualizations.

1. A network graph to show the institutional ownership network, with node size representing firm centrality and edges representing shared ownership.
2. A histogram or density plot to show the distribution of return correlations during stress periods, so visualize how much stocks tend to move together when markets are volatile.
3. A scatter plot to show the relationship between ownership overlap and return correlation for firm pairs.

The timeline is as follows.

1. By the end of Week 5
  - a. Figure out the definition of market stress (daily stress days vs. multi-day stress periods) using the realized volatility data already collected from FRED
  - b. Clean and finalize the stress\_days.csv file
  - c. Finalize the sample of mega-cap firms using Yahoo Finance data
2. Week 6
  - a. Use the Python interface to WRDS to collect daily stock return data for firms
  - b. Clean and merge stock return data with stress indicators
  - c. Check and clean return data (missing values, alignment across firms)
3. Week 7
  - a. Use the Python interface to WRDS to collect institutional ownership (13F) data
  - b. Construct the firm-level institutional ownership network
  - c. Compute network measures
  - d. Validation checks on the network
4. Week 8 before Feb 23
  - a. Pairwise stock return correlations during the decided upon stress periods
  - b. Conduct descriptive analyses linking ownership overlap to return comovement
  - c. Produce visualizations